Comparing Support Vector Machine (SVM) and Multilayer Perceptron (MLP) in predicting the risk of heart attack

Abstract

This study compares Support Vector Machine (SVM) and Multilayer Perceptron (MLP) models for predicting heart attack risk using clinical indicators from the Mendeley Heart Attack dataset. Through feature engineering we identified Pulse Pressure, Age, CK-MP and Troponin to be key features. Both optimized models achieved 95% accuracy on the test set, with the MLP showing slightly higher discriminative power (AUC 0.97 vs 0.93).

1. Introduction

Heart disease, including heart attacks, are among the leading causes of death worldwide. According to WHO, cardiovascular diseases take the lives of approximately 17.9 million annually. Within that alarming number, the majority is caused by heart attacks [1]. Despite major progress in the medical fields, subtle warnings that may cause heart attack are ignored even though these signs can be seen from lab results and vital recordings. Using these in indicators in combination could unravel insights valuable for patient's health.

In this paper, the performance of two models is evaluated and compared in predicting the likelihood of suffering from a heart attack based on a ranged of clinical indicators. The models considered for analysis are Support Vector Machine (SVM) and Feedforward Multilayer Perceptron (MLP). Different configurations of these models are trained and optimised to tackle the issue at hand.

In Section 2, the dataset used for analysis and modelling is introduced and briefly explained. Section 3 covers the methods used for training, evaluating and improving the models. Section 4 consists of the choice of parameters and experimental results and in Section 5 we present a fair comparison and findings. Section 6 concludes the paper.

1.1 Support Vector Machine (SVM)

SVM is a powerful supervised ML algorithm that identify the optimal hyperplane that maximises the margin between classes in a dataset. Data are transformed into higher dimensions using kernels to handle non-linear relationships, and as mentioned by Cristianini and Shawe-Taylor (2000), The kernel trick enables SVMs to implicitly work in higher dimensions by computing only inner products, making it computationally efficient compared to explicit coordinate calculations [2]. SVMs are memory efficient as they use a subset of training points (support vectors) for the decision boundary, but they can be computationally intensive for large datasets and may require proper tuning for hyperparameters like kernel choice and regularization.

1.2 Multilayer Perceptron (MLP)

MLP is an artificial neural network (ANN) which composes of an input layer, one or more hidden layers and an output layer and it is trained using backpropagation. Backpropagation allows the network to identify important patterns in the data and hence enabling accurate predictions on unseen data; the network adjusts connecting weights through repeated iterations where it compares predictions to actual values and create an error signal that is backpropagated through the network [3].

MLP can model non-linear relationships and can learn complex patterns through the layered architecture using activation functions however without proper regularization and hyperparameter tuning, there can be overfitting. MLPs are also sensitive to initial weights and biases which can lead to variations in performance.

2. Dataset

The dataset used for analysis is the heart attack dataset from Mendeley, which was collected at Zheen Hospital in Iraq [4]; it contains clinical measurements that are essential for cardiovascular health assessment. There are 1,319 instances and 9 variables. Table 1 shows the summary statistics of medical features, and it observed that patients with positive results tend to older (mean age of 59). Cardiac biomarkers like CK-MB (a creatine enzyme) are significantly higher in positive cases; Troponin (a protein found in heart muscles) also shows similar patterns. Blood pressure measurements and heart rates show less dramatic differences between the groups, suggesting these vital signs alone may be less discriminative for diagnosis. The dataset provides a balanced foundation for evaluating pattern recognition algorithms, with 810 positive cases and 509 negative cases.

Table 1: Summary Statistics of Medical Features

	Positive (810)				Negative (509)			
Features	**Mean**	**Std dev**	**Min**	**Max**	**Mean**	**Std dev**	**Min**	**Max**
Age	59	12.96	19	103	52	13.7	14	91
Gender	0.69	0.46	0	1	0.61	0.49	0	1
Heart rate	78.62	53.69	20.0	1111	77.89	48.21	20.0	1111
Systolic Blood Pressure	126.74	25.54	65.0	223	127.86	27.04	42.0	223
Diastolic Blood Pressure	72.16	13.86	38.0	154	72.44	14.33	40.0	128
Blood sugar	144.67	72.63	35.0	541	149.76	78.41	60.0	541
CK-MB	23.27	57.70	0.353	300	2.56	1.37	0.32	7.02
Troponin	0.571	1.39	0.003	10.3	0.027	0.44	0.001	10.0

2.1 Exploratory Data Analysis

Figure 1: Heart Attack Data Analysis

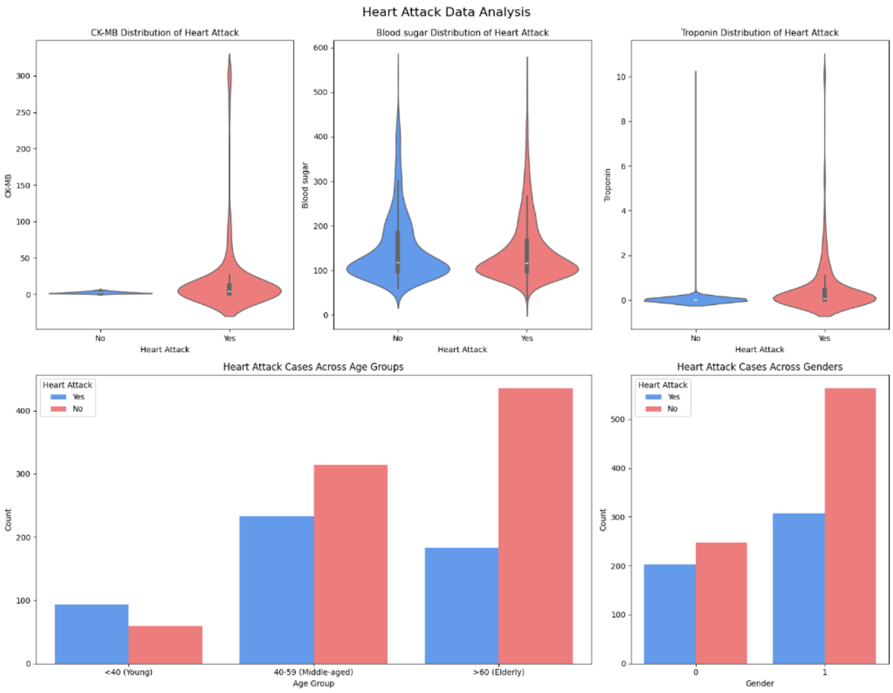


Figure 1 shows a combination of graphs as part of initial analysis. The violin plots (top row) show the distribution of important biomarkers. CK-MB and Troponin displays higher levels in heart attack cases (red) compared to non-heart attack cases (blue). These biomarkers are particularly important as they are specific indicators of cardiac damage. Blood sugar is slightly lower in heart attack cases.

The bottom row shows demographic patterns. Age distribution shows that heart attack cases increase with age (above 60 experiencing the highest rate). Heart attack cases are also prevalent in middle aged group. Gender distribution demonstrates that males (gender = 1) experience a considerable higher heart attack rate than female (gender = 0). From EDA, we identified that CK-MB and Troponin are primary indicators of heart attack and age adds important supplementary information. Blood sugar may not yield much predictive power. These observations suggest that feature selection focusing on cardiac biomarkers and demographic factors would be crucial for building effective classification models.

2.2 Feature Engineering

Through correlation analysis, it was observed that Systolic blood pressure and Diastolic blood pressure had a strong positive correlation of 0.59 with each other, but neither showed strong relationship with the target variable. This observation led to the creation of a new feature called Pulse Pressure; and the formula is shown below, where systolic blood pressure is the maximum pressure that your heart applies when beating and diastolic blood pressure is a measurement of the pressure in your arteries between heartbeats [5].

$$Pulse\ Pressure = Systolic\ Blood\ Pressure - Diastolic\ Blood\ Pressure$$

Previous studies have also revealed that Pulse Pressure is a major indicator of the onset of heart related problems, it represents the force your heart generate for each contraction [5].

3. Methodology

The dataset is split into 80:20 ratio for training and testing (1055 Training: 264 Testing). A stratified 10-fold cross validation is applied on the training set; this ensure that all the available data is used on training and validation and is suitable in our case as a minor class imbalance was observed in the dataset [6]. During training, models are compared using mean accuracy.

A base model for SVM and MLP are initially ran with all available features. Permutation Importance is applied to each model to identify features with high predictive score and the models are trained with a reduced set of features to see if performance is affected. In fact, Permutation importance evaluates the decrease in a model's performance when a feature's values are randomly shuffle [7]. Hyperparameter tuning is then performed using Grid Search.

For the MLP model, we implement both Sci-kit Learn library and PyTorch framework to compare performance however grid search is only done with the Sci-kit Learn version. A variety of hyperparameters have been investigated including early stopping.

Algorithm comparisons involved evaluating both models on the same holdout test set using accuracy, precision, recall, and F1-score metrics, which provided a fair and comprehensive assessment of their relative strengths in classifying heart attack cases.

4. Choice of parameters and Experimental Results

4.1 Support Vector Machine (SVM)

For the base SVM model, RBF (Radial Basis Function) has been selected since EDA showed non-linear relationships between the features and target variable. Linear separation is unlikely to perform well given the complex interactions of biomarkers with heart attack. To provide a balanced trade-off between maximising the margin and minimizing error, the regularization parameter (C) is set to 1; this should mitigate overfitting. The gamma parameter is set to 'scale' to automatically adapt to variance in the dataset. This configuration results in an accuracy of 66.6%.

We subsequently performed Permutation Importance to identify the most important features. The number of features has been reduced from seven to five. The new set of features includes: 'Age', 'Pulse Pressure', 'Blood sugar', 'CK-MB' and 'Troponin; and the model with the reduced set of features results in 78.9% accuracy. Further reduction in the set of features has been explored but lower accuracy has been observed.

The next step consists of Grid Search using stratified 10-fold cross validation; where table 2 shows the hyperparameters tuned. The Grid explores regularisation (C), 0.1 to 100, to test whether strong penalty on misclassification might help in predicting heart attack, where false negatives could be fatal. Both linear and RBF kernels are included to evaluate and confirm our initial assumption about non-linearity. The Grid Search identified the optimal configuration as C=100, gamma=0.1 and the RBF kernel, achieving a cross-validation accuracy of 95.93%. The high C value in the optimal model indicates that minimizing misclassification errors was more important than maximizing the margin width for this dataset, suggesting that a more complex decision boundary is important to capture underlying patterns.

Table 2: Parameter Grid for SVM

Regularisation (C)	0.1, 1, 10, 100
Kernel	'rbf', 'linear'
Gamma	'scale', 'auto', 0.1, 0.01

4.2 Multilayer Perceptron (MLP)

The base MLP model, we initially use **Scikit-Learn**'s default 'MLPClassifier' configuration. This consists of one hidden layer with 100 neurons, 'ReLU' activation function and the 'adam' optimiser for backpropagation with a learning rate of 0.0001. There is also L2 regularisation (weight decay) which penalises large weights. Maximum iterations are 200 but this led to convergence warning, indicating that more iterations are needed. Despite these constraints, a relatively high accuracy of 75.86% is observed.

During feature selection, it is found that CK-MB and Troponin are impactful features especially for MLP. With the new set of features (as seen above), the MLP still fails to converge, however accuracy increases unexpectedly to 94.70%.

MLP Grid Search is also done using 10-fold stratified cross-validation. Table 3 shows the hyperparameters tuned for MLP. Several neural networks architectures have been tested with different hidden neurons and layer configurations. We use ReLU and Tanh to evaluate their impact on non-linear data. Alpha varies to identify optimal generalisability. More notably is the number for maximum iterations, we use values up to 1005 to address the convergence warnings. Early stopping help with overfitting. The Grid Search concluded with an optimised model with two hidden layers (first layer with 20 neurons and second layer with 10 neurons), ReLU activation function, 'adam' optimiser and relatively strong regularisation of 0.01. This generated an accuracy of 96.60%.

Table 3: Parameter Grid for MLP

Hidden layer(s) and neurons	(20) ;(50) ;(21,10) ;(50,20)
Activation function	ReLU, Tanh
Optimiser (Backpropagation)	'adam', 'sgd'
Alpha (L2)	0.0001, 0.001, 0.01
Learning rate	0.001, 0.001
Max iterations (epoch)	500, 750, 1005
Early stopping	True

A **PyTorch** version of the MLP is also explored. For the base model, we configure the neural network to match the Sci-kit learn 'MLPClassifier' as closely as possible. The PyTorch implementation offers greater flexibility and control. A custom network with forward propagation with one single hidden layer with 100 neurons and ReLU activation. Loss calculation is initialised using cross-entropy along with backpropagation. The implementation utilized mini-batch processing with a batch size of 32 to balance computational efficiency with gradient stability. This version being more tedious to implement that sci-kit learn has no convergence issue and surprisingly generate high accuracy. The base PyTorch model has an accuracy of 93.2%. With the set of less features, we observed even higher accuracy of 93.9%.

We have not performed Grid Search specific to the PyTorch model, but we have used the optimised hyperparameters obtained from the Grid Searchof the best Sci-kit learn model. The PyTorch model with the optimised hyperparameters resulted in lower accuracy of 96.5%. A Grid Search tailored for the PyTorch model could improve the model to rival Scikit Learn.

5. Results, Findings and Evaluation

From the confusion matrices in Figure 2, clear improvements can be seen from the baseline models to the optimised models for both algorithms. Both best models achieve identical performance in the test set. Compared to the baseline model, a decrease in misclassification of false positives (47 to 6), however the misclassification of false negatives increases from 4 to 8. The best MLP model and base model show similar false negatives while a decrease in false positives is observed in the best model. In terms of the prediction of true values, both models show improvements. The ROC curves in Figure 3 quantify this performance difference with AUC values of 0.93 for SVM and 0.97 for MLP, suggesting the neural network approach has a slight edge in classification power. The classification reports of both SVM and MLP achieved identical overall performance. Both have 95% accuracy on the test data.

Figure 2: Confusion Matrices

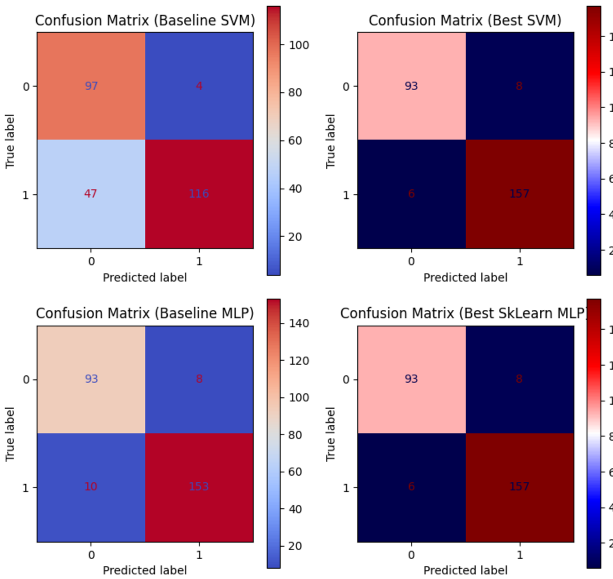
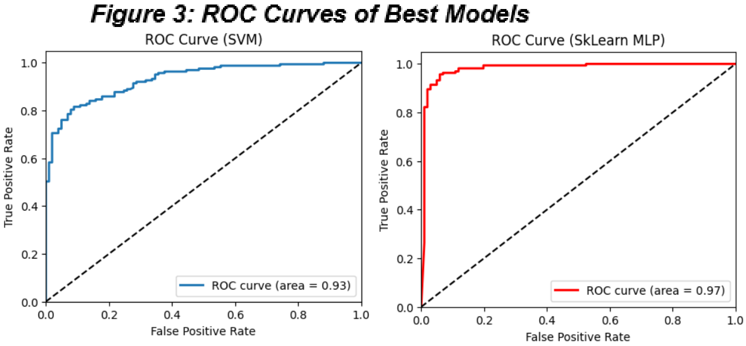


Figure 3: ROC Curves of Best Models



In terms of the architecture of the models, SVM performed well with a reduced set of features and the RBF kernel. It captures non-linear decision boundaries with high regularization value (C=100), indicating that the model prioritised minimising misclassifications over a larger margin width. MLP achieved better performance with a relatively simple architecture of two hidden layers with 20 and 10 neurons. The ReLU activation function and regularisation (alpha=0.01) most likely contributed to the capture of complex relationship between the features and target variable. However, the high C value in SVM could potentially lead to overfitting and with MLP, we faced some challenges with convergence.

Through feature important, we identified that age, CK-MB and Troponin are important features for both models, which aligns with medical knowledge since those indicates abnormalities in heart health. SVM placed higher importance of age while for MLP, CK-MB was the most vital.

Both optimised models have the potential to be viable tools in assisting in diagnosing risk of heart attacks. The models' strong performance in correctly identifying positive cases (96% recall for both models) is particularly valuable in emergency medicine, where missing a heart attack diagnosis can have life-threatening consequences. SVM would offer greater interpretability while MLP would be more flexible to new patterns that develop in future data. The smaller set of features makes the models more practical. By requiring fewer biomarkers for accurate prediction, healthcare providers could reduce testing costs and accelerate diagnosis times. Both models would require similar computational resources for implementation.

The findings from our analysis and modelling compare favourably with recent studies in the field. El-Sofany et al. (2024) reported 97.57% accuracy is reported using machine learning algorithms which is similar to the MLP approach both Scikit Learn and Pytorch [8]. Audrey and Widianto (2023) reported 85.53% accuracy with SVM models for heart attack prediction, which is lower than our results. Their finding that chest pain type was the most important feature aligns with our feature importance analysis, though our research identified additional significant biomarkers [9].

6. Conclusion

The techniques implemented for both SVM and MLP (PyTorch and Scikit Learn) followed proper cross validation and effective hyperparameter tuning was used through Grid Search. Using comparable performance metrics like accuracy, we identified the best models for out dataset. However, the simple MLP architecture may not capture more complex relationships; introducing new features may cause redundancy and SVM would struggle with larger datasets; mini batch training should be considered. The Grid Search for MLP was computationally expensive

The use of PyTorch, despite taking a long time to set up, should be explored further; the model generated high performance on the base model and there is the benefit of using pre-trained models. The trade off between the greater flexibility of PyTorch against Scikit Learn simple implementation and SVM ease of interpretation should be considered when deciding on the implementation.

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